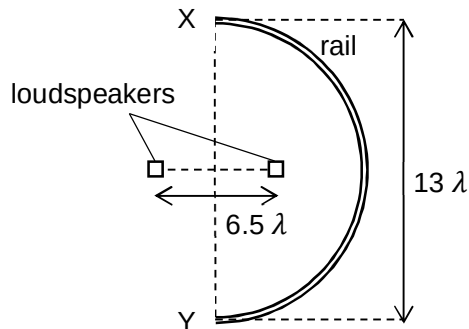


- 19 Two loudspeakers are emitting sound of wavelength λ in all directions. They are in phase with each other and are placed a distance 6.5λ apart in the middle of a semicircular rail of diameter 13λ , as shown below. Moveable microphones along the rail are used to detect the sound intensity along the rail. The midpoint of the line joining the 2 speakers coincides with the centre of line XY.



How many minima will the microphones detect?

- A 7 B 12 C 13

Ans : C

At the centre of the rail, the path difference is 6.5λ .

At the start and end of the rail, the path difference is 0λ .

Hence, the magnitude of path difference increases from 0 to 6.5λ and then decreases to 0

Since waves are in phase, minima will be detected at locations where path differences are $(n+1/2)\lambda$ and this occurs at path differences = 0.5λ , 1.5λ , 2.5λ , 3.5λ , 4.5λ , 5.5λ , 6.5λ .

- D 14

20. A laser light of wavelength 650 nm is passed normally through a narrow slit. A screen is placed